



Environment Services

EXL

Team A

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Environment Services Voice Agent

Automated Customer Service & Pickup Management Using Conversational AI

The Problem: Manual Operations at Scale

This Agent providers face mounting pressure from rising call volumes, missed pickups, and inefficient manual processes — driving up costs and eroding customer trust.

Missed Pickups

No automated notification or resolution path for customers.

Manual Support

High call center workload with long wait times and inconsistent responses.

No Real-Time Status

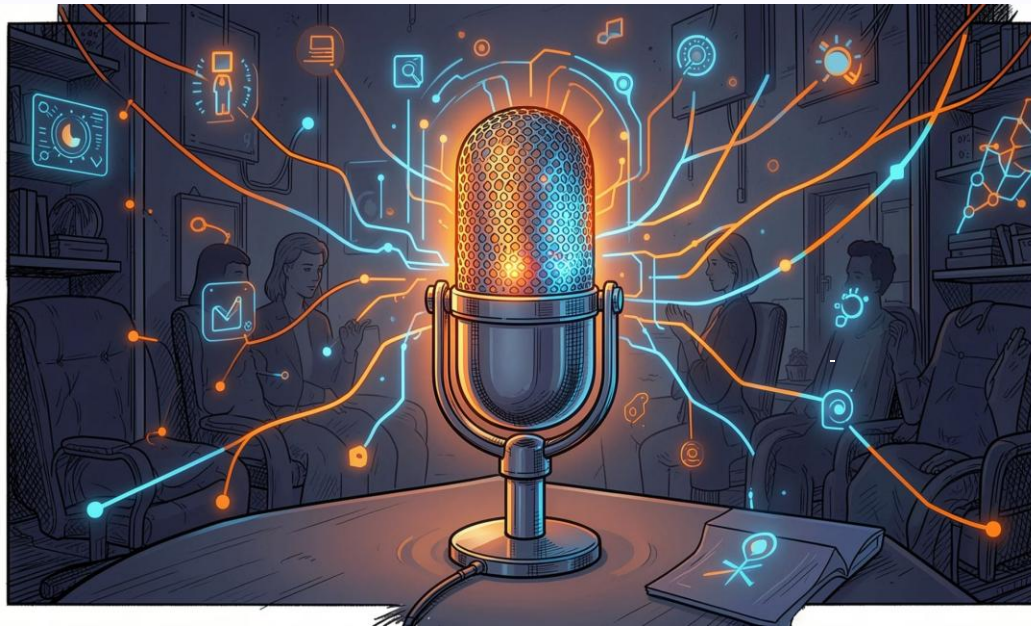
Customers lack visibility into service schedules or delays.

High Operational Costs

Inefficient rescheduling and escalation processes inflate overhead.



Solution Overview: The AI Voice Agent



An intelligent conversational AI system that handles end-to-end customer interactions — from authentication to issue resolution — without human intervention.

- **Authenticate Users**
Secure identity verification with retry logic and escalation.
- **Handle Service Requests**
Missed pickup reports and rescheduling handled automatically.
- **Provide Status Updates**
Real-time service information delivered instantly.
- **Escalate When Needed**
Complex issues routed seamlessly to human agents.

System Objectives

Six core goals drive the design of the AI Voice Agent — balancing customer experience with operational efficiency.



24×7 Support

Always-on availability, eliminating wait times and after-hours gaps.



Automate Requests

End-to-end automation for routine service interactions.



Reduce Costs

Lower call center volume and manual processing overhead.



Improve CX

Faster resolution and consistent responses boost satisfaction.



Operational Efficiency

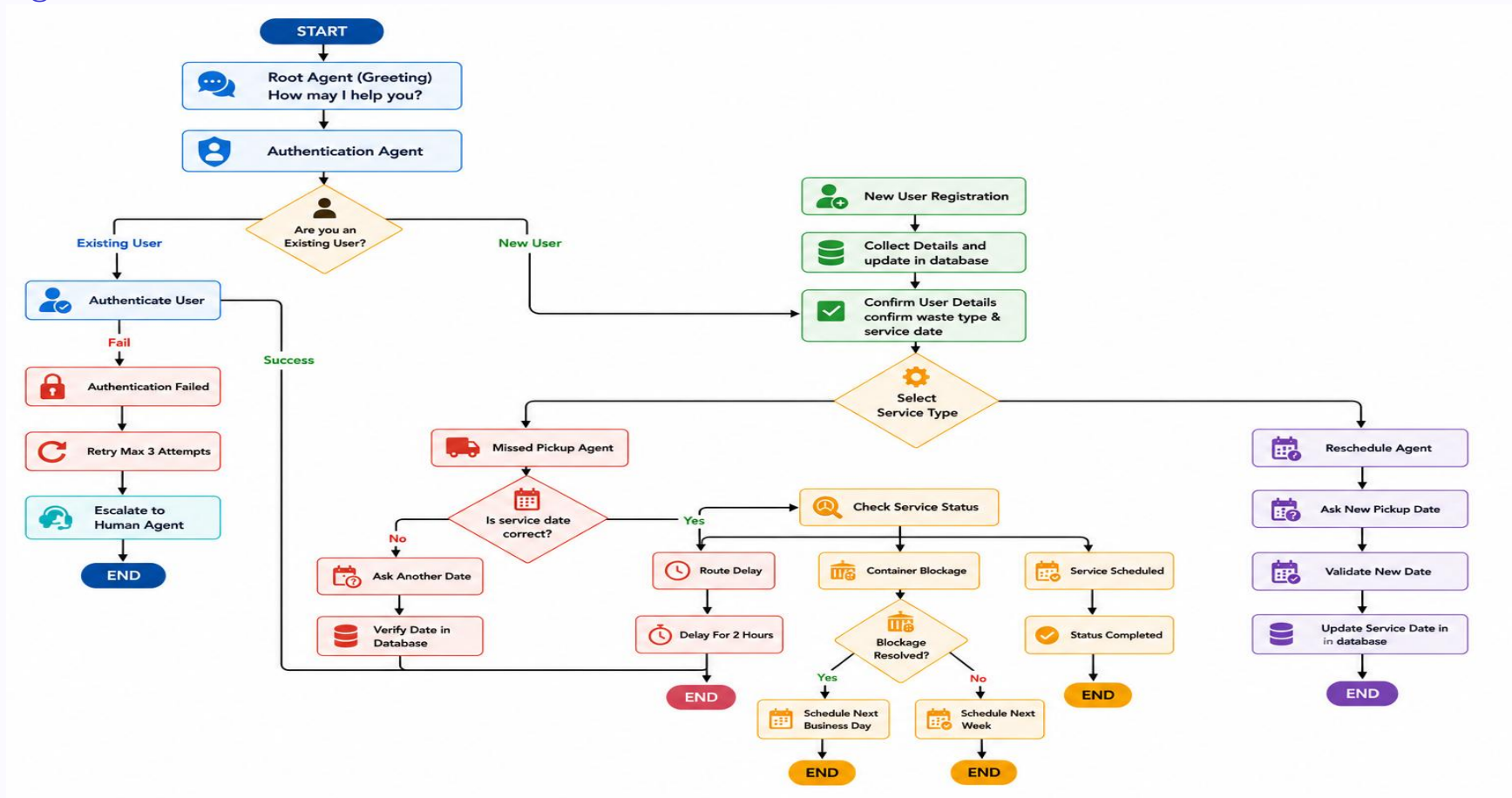
Streamlined workflows reduce agent workload and errors.



Real-Time Tracking

Live service status updates delivered directly to customers.

Agent Workflow



Authentication & Registration Flows

Existing User Authentication

- User provides identifying information
- System verifies credentials against database
- Success → Proceed to service selection
- Failure → Up to 3 retry attempts
- After 3 failures → Escalate to human agent

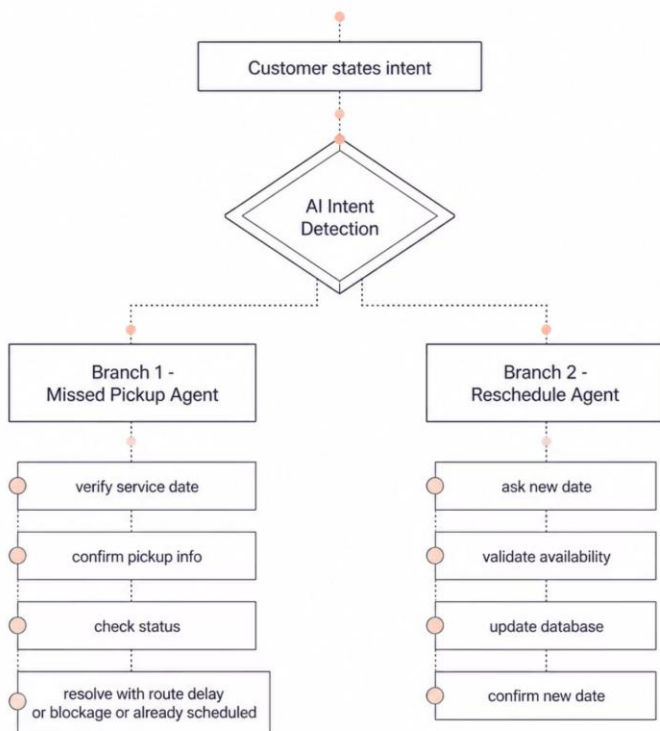
New User Registration

1. Initiate new user registration
2. Collect name, address, and contact details
3. Update backend customer database
4. Confirm service type and coverage area
5. Confirm and schedule first pickup date

📄 Registration data syncs in real time with the waste management CRM.

Service Selection & Missed Pickup Resolution

The AI detects customer intent through natural language and routes the call to the appropriate specialized agent.



Missed Pickup Outcomes

Route Delay

Inform customer of ~2-hour delay. Request closed automatically.

Container Blockage

Resolved → Next business day. Unresolved → Schedule next week.

Already Scheduled

Notify customer of existing pickup date. No action needed.

Reschedule Agent & Technical Architecture

Pickup Rescheduling Process

01

Request New Date

Agent asks customer for preferred pickup date.

02

Validate Availability

System checks route capacity and service calendar.

03

Update Database

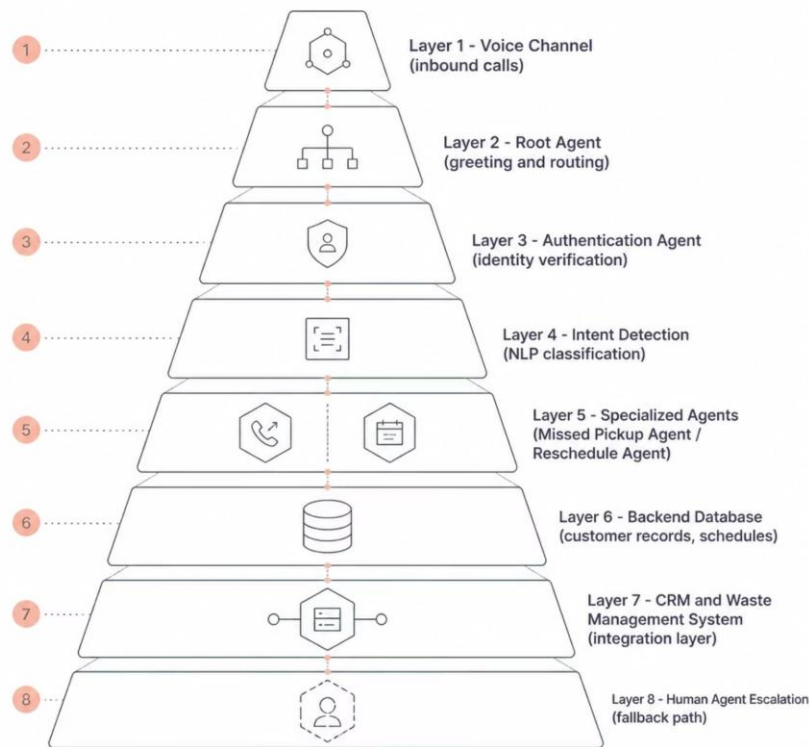
New date committed to CRM and routing system.

04

Confirm to Customer

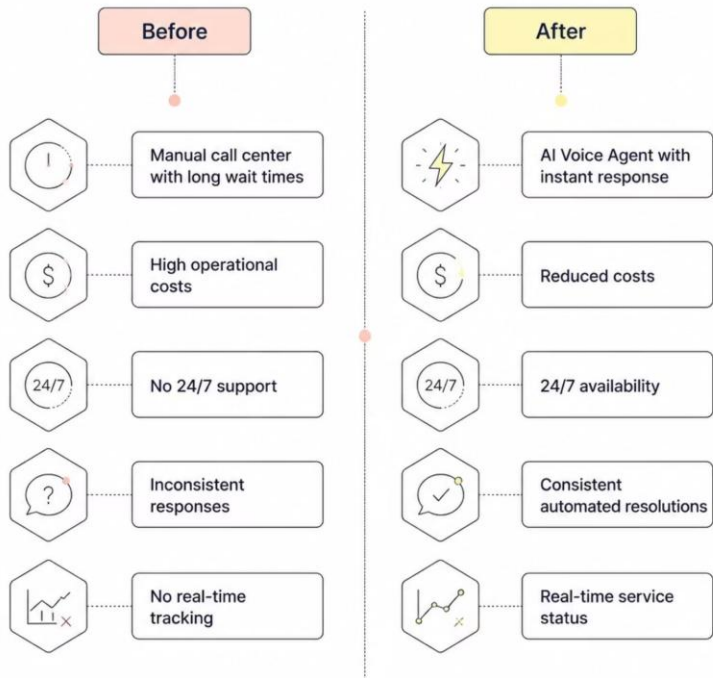
Verbal confirmation with date, time, and reference ID.

System Architecture



Business Impact & Benefits

The AI Voice Agent delivers measurable improvements across cost, efficiency, and customer satisfaction — transforming a manual, reactive operation into an automated, proactive service model.



Key Outcomes

- **24x7 Availability** — No after-hours coverage gaps
- **Reduced Call Volume** — Automation handles routine requests
- **Faster Resolution** — Instant responses vs. manual processing
- **Lower Costs** — Reduced agent workload and overhead
- **Higher Satisfaction** — Consistent, reliable customer experience

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Thank You